Review

Locking and Low Energy Holding in Soft and Hybrid Robotic Grippers

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Abstract

Soft grippers can conform to fragile objects, but maintaining a grasp may require continued actuation or an additional retention mechanism. Claims of low energy holding remain difficult to compare because mechanical locking, trapped pressure, and reduced control power describe different system properties. This critical narrative review examines self-locking transmissions, bistable structures, jamming, material fixation, and selected adhesive and field-based interfaces. A two-axis framework separates the physical load-retention mechanism from its dependence on sustained supply, isolated resources, or state-retained conditions. Published examples show that eliminating actuator input during holding does not establish zero system energy, reliable retention over time, or economical release. We therefore compare evidence through load paths, resource boundaries, holding duration, and release requirements rather than ranking unmatched payload or energy values. An explicit cycle-energy balance identifies when transition costs can outweigh holding savings, while a reporting checklist connects retention tests with power interruption, force relaxation, and release under load. The resulting synthesis supports mechanism selection for prolonged holding and contact-triggered manipulation, while identifying the measurements needed before energy-saving claims can be transferred between grippers.

Keywords: soft robotic grippers; self-locking; bistability; jamming; grasp retention; energy consumption; release mechanisms; variable stiffness

1. Introduction

A soft gripper may close successfully and still be unsuitable for a prolonged holding task. Conforming fingers establish contact, but the resulting grasp also needs a load path that resists slip, opening, and deformation. Soft materials and compliant architectures enable useful adaptation, while their deformation complicates force transmission and control [1, 2]. The comparison should therefore extend beyond contact formation to the interval during which the object must remain secured.

Actuation, stiffness adjustment, and retention are related functions with different requirements. An actuator moves a finger; a stiffening element changes its mechanical response; a retaining element prevents loss of the required grasp state. One component may perform several functions, but the functions are not interchangeable. Reviews of actuation and stiffening provide extensive accounts of their physical principles and design trade-offs [3, 4]. Those accounts do not make stiffness enhancement alone evidence of low energy holding.

The distinction matters when interpreting descriptions such as passive, self-locking, electronics-free, and source-free. A contact-triggered mechanism can avoid an electronic command while consuming stored elastic energy. A valve can preserve pressure after a pump stops, yet the grasp still depends on that pressure. A latched joint may retain its angle while the soft contact relaxes. These examples motivate separate questions about energy input, resource persistence, and object retention.

Several existing reviews overlap substantially with this topic. Broad gripper reviews organize material systems, actuation, stiffness, adhesion, and applications [5, 6]. A somatic-design review examines interactions among gripper functions [7]. Kim et al. integrate design, actuation, and sensing, including thermal shape locking and its energy and durability trade-offs [8]. Reviews of bistable actuators, jamming,

and laminar locking provide detailed mechanism accounts [9, 10, 11, 12]. The present contribution is therefore a focused comparison of retention conditions, rather than another general technology inventory.

The recent review of passive adaptive grippers is especially relevant because it distinguishes externally sustained from internally sustained retention [13]. That distinction is an important antecedent of the present resource analysis. Our scope extends across actively closed and passively triggered soft and hybrid grippers. We connect their retention mechanisms to energy accounting, source interruption, contact-force persistence, and release under load. The contribution is this combined comparison and its evidence requirements, rather than the invention of passive retention as a concept.

The central question is which physical mechanism retains the load, with what continuing resource dependence, and at what cost of release and reset. We address it through a two-axis classification, a condition-aware comparison of representative studies, and a cycle-energy model. We also propose a minimum reporting set for future experiments. The guidance addresses mechatronic design and robotic manipulation, including controller decisions about acquisition, engagement, holding, and release. Table 1 positions this combined comparison against the selected earlier reviews.

Table 1. Positioning against selected existing reviews. The final column states the additional question pursued here; it does not imply that the earlier review omitted every related discussion.

Existing review	Primary organizing focus	Relationship to the present review
Shintake et al. [2]	Actuation, controlled stiffness, and controlled adhesion	Provides the broad technology map; here the unit of comparison is a completed retention-and-release task.
Chi et al. [9]	Bistable elements, materials, and robotic functions	Provides structural principles; here stability is distinguished from load-bearing grasp retention.
Fitzgerald et al. [10]	Granular, layer, and fibre jamming	Provides jamming taxonomy and testing gaps; here confinement is linked to supply isolation and hold duration.
Caro and Carmichael [11]	Laminar lock/unlock mechanisms and applications	Provides mechanism detail; here release and energy boundaries are compared across retention families.
Granular-gripper review [12]	Design, control, and applications	Provides a dedicated gripper account; here jamming is compared with latches, bistability, and material fixation.
Passive adaptive grippers [13]	Passive acquisition, retention, and release	Provides the sustained-retention distinction; here it is connected to active soft grippers and complete-cycle energy.
Kim et al. [8]	Integrated morphology, actuation, sensing, and variable stiffness	Already discusses thermal locking and energy trade-offs; here the comparison follows resource interruption, persistence, and release across mechanisms.

2. Review Approach and Evidence Boundaries

This article is a critical narrative review based on targeted, iterative retrieval completed on 19 October 2026. It is not a systematic review or meta-analysis. The search used a general web search interface, publisher and institutional pages, Crossref bibliographic records, PubMed Central, and arXiv. Searches combined soft gripper or hybrid gripper with self-locking, bistable, jamming, shape locking, low power, low energy, holding, and release. Exact-title and DOI searches followed promising records. Reference lists in retrieved reviews were used to locate relevant primary studies.

Studies were retained when they described a physical retention mechanism, informed its release or energy dependence, or supplied essential review context. Purely rigid grippers and generic actuator demonstrations were peripheral unless they clarified a mechanism used in soft or hybrid grasping. Hybrid devices were included when compliant contact or adaptation remained functionally important. Adhesion and magnetic or

electrostatic systems serve as boundary cases where low electrical input can be mistaken for autonomous mechanical retention.

The evidence ledger records each identifier, mechanism, consulted source, access level, supported statement, and extraction limit. Accessible full texts were preferred. Publisher-deposited abstracts were used for restricted claims when full text was unavailable; those entries are explicitly identified. Search-result descriptions alone were not treated as proof of a quantitative result. Bibliographic matching used title and authors as well as DOI, because a resolving identifier can still belong to the wrong paper.

Quantitative values were retained in their reported context. We did not pool maximum payload, pull-off force, stiffness ratio, electrical switching energy, or cycle energy into a common ranking. Missing information is recorded as not established from the consulted source, rather than assumed absent from the original study. No artificial screening totals, duplicate counts, independent-reviewer procedure, or statistical synthesis are reported. The supplementary search log documents the actual queries and access failures, which constrain completeness.

Recent archival manuscripts were checked for later publication information where available. The source-free fluidic gripper has an arXiv version revised in April 2026 and identifies a RoboSoft 2026 conference version [14]. The ratchet-gripper manuscript identifies acceptance to ICRA 2026 [15]. We cite the versions actually inspected and distinguish their archival records from independently verified proceedings metadata. Their energy-related statements receive the same boundary checks as older journal articles.

3. A Framework for Retention and Resource Dependence

The first axis identifies the dominant physical load-retention mechanism. Positive engagement constrains motion through teeth or geometric stops; self-locking transmissions resist backdriving through geometry and friction. Elastic bistability retains a configuration within a loaded equilibrium basin. Frictional confinement suppresses relative motion between particles, layers, fibres, or contacting surfaces. Material fixation preserves geometry through a stiff material state. Adhesive and field-based interfaces transmit load across the contact itself. These are functional families that may overlap within a single gripper, rather than mutually exclusive material classes.

The second axis describes the resource condition at a named retaining element. Supply-sustained retention requires applied drive, maintained electrical bias, or replenished fluid input. Resource-isolated retention depends on fluid pressure or electric charge trapped after isolation. State-retained operation instead relies on a mechanical constraint, loaded elastic equilibrium, solid material state, or remanent magnetization, without continuing drive or pressure/charge support at that element. These operational labels distinguish dependencies, not all forms of stored energy; elastic energy remains relevant in a state-retained structure. Mere connection to an inactive supply does not determine the label. When the source condition is unspecified, the classification remains unverified.

The axes prevent several category errors. Vacuum jamming identifies confinement, but not whether the pump runs or the chamber is isolated. Bistability identifies an energy landscape, but not whether its loaded state supports the object. Remanent magnetization may persist without coil current while an associated fluid circuit remains powered. A hybrid gripper can therefore require several component-specific descriptions. In a closed hydraulic bistable gripper, structural equilibrium and contained liquid jointly support operation; absence of an external pump does not remove the contained fluid from the load path [14].

Four boundaries should be recorded alongside the classification: the retained variable, the disturbance envelope, the holding interval, and the release condition. Retained variables include joint position, actuator shape, contact force, pressure, and object pose. A position lock only directly constrains some of these variables. Evidence of object retention must specify which variables were measured or observed, and how unacceptable motion or force loss was defined.

For an illustrative conservative, single-coordinate bistable mechanism, equilibrium under a constant generalized load f can be described by the total potential $\Pi(q; f)$. Local stability requires

$$\frac{\partial \Pi}{\partial q} = 0, \quad \frac{\partial^2 \Pi}{\partial q^2} > 0 \tag{1}$$

These conditions apply at the loaded equilibrium, rather than only to the unloaded actuator. Object contact can shift the equilibrium, change the energy barrier, or prevent access to the intended closed state. Frictional and unilateral-contact systems require additional constraint and dissipation analysis; Equation (1) is not a general stability test for jamming or ratchets.

A practical retention claim should consequently answer three distinct questions. Does the gripper resist the specified load after its stated source is interrupted? Does that capability persist for the intended duration? Can the object then be released through the intended command or interaction? Figure 1 organizes these questions around a complete grasp cycle. Table 2 connects the framework to the dominant failure and release modes.

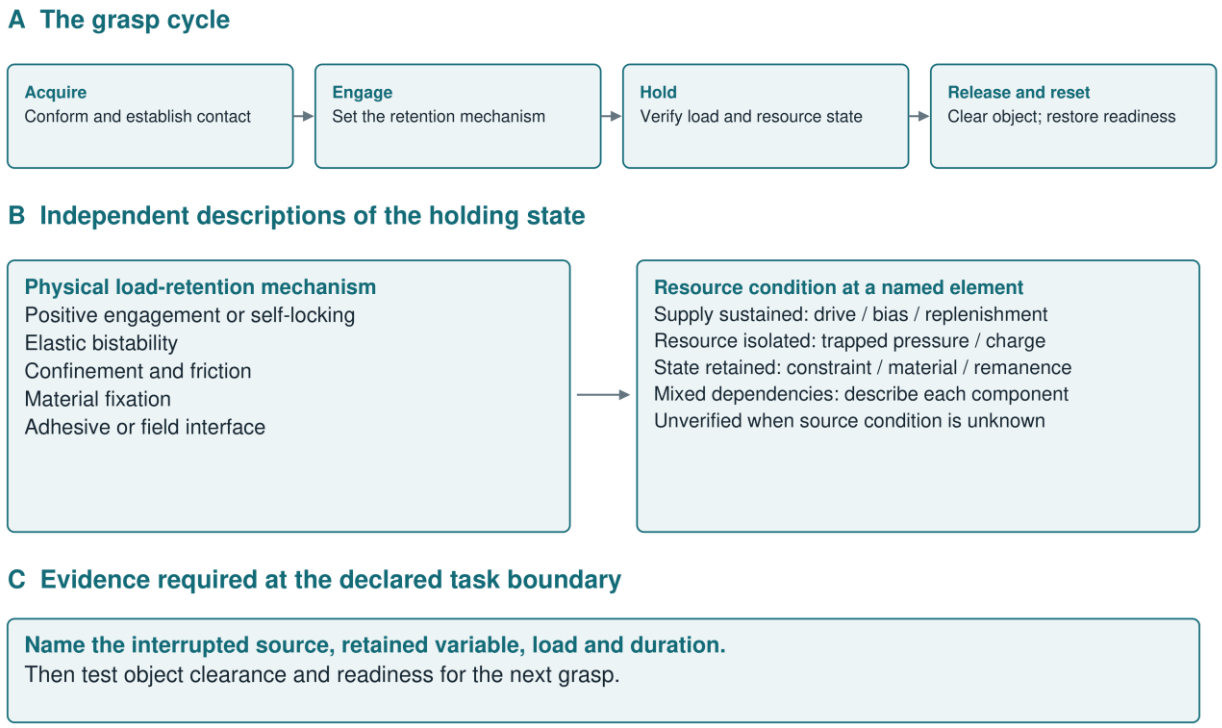


Figure 1. Retention assessment across the grasp cycle. The physical load path and the resource condition are separate axes. The interruption test must name the disconnected source and the variables still supported. The diagram is a conceptual synthesis, not an experimentally validated controller.

Table 2. Mechanism-level comparison. Resource conditions are conditional on the implemented circuit and test. Listed failure modes are design considerations, not measured failure rates.

Dominant mechanism	What retains the load	Resource and release implications	Principal evidence needed
Positive engagement / self-locking transmission	Engaged teeth, geometric stop, or resistance to backdriving [16, 17]	Mechanical holding is possible; disengagement may require actuator effort or prior unloading.	Backdrive or tooth capacity; force relaxation at fixed position; release under the retained load.
Elastic bistability	Loaded equilibrium and an energy barrier [9, 18, 19]	No sustained drive is needed for an adequate stable state; reset supplies work.	Load-dependent stability, inadvertent triggering, capture losses, and reset requirements.
Confinement and friction	Normal stress increases resistance to internal slip [20, 21, 22]	Vacuum, pressure, mechanical clamping, or fields may supply confinement; venting or unclamping releases.	Confinement persistence; leakage or creep; slip and object-interface failure.

Dominant mechanism	What retains the load	Resource and release implications	Principal evidence needed
Material fixation	A stiff solid or glassy state constrains deformation [23, 24, 25]	Holding may persist after cooling; heating and sometimes active cooling affect cycle energy.	Operating temperature, dwell time, thermal drift, and release or reconfiguration time.
Adhesive or field interface	Interfacial traction, sometimes combined with shape fixation [26, 27, 28]	Detachment can require peeling, unloading, reheating, or field removal; low current does not prove source independence.	Surface dependence, residual adhesion, field or temperature condition, and release completion.

4. Physical Mechanisms and Their Holding Limits

4.1. Positive Engagement and Self-Locking Transmissions

Mechanical locks separate compliant acquisition from a stiffer holding load path. A finger first conforms through pneumatic or tendon actuation, and engagement subsequently limits reopening. Guo et al. describe an exoskeleton with ratchet-and-pawl locking and a pneumatic unlocking actuator [16]. Their publisher-deposited abstract reports a load capacity up to 9 kg and holding without compressed gas. These statements support a state-retained mechanical architecture, but the abstract alone does not establish a long-duration force record or matched full-cycle energy.

Self-locking can also arise in a transmission rather than a discrete latch. Tang et al. use a self-locking screw to coordinate a variable-stiffness soft gripper [17]. The design illustrates why the motor that creates motion and the mechanism that resists backdriving should be described separately. Resistance to backdriving remains conditional on geometry, friction, lubrication, wear, and load direction. The term self-locking should not substitute for a measured load and disturbance envelope.

Engagement offers a direct mechanical route to removing holding input, but introduces discrete constraints into an otherwise adaptive structure. Tooth pitch or backlash can change the final finger angle, and engagement can redistribute force between contacts. Once the joint is locked, an elastomer or a soft object may still relax. A design that conserves joint position therefore needs separate evidence that contact force remains adequate.

Recent work continues to develop soft-rigid coupling rather than treating compliance and load-bearing as mutually exclusive. Xiao et al. report separate actuation and self-locking modes in a rigid-soft bending actuator [29]. The available abstract supports improvements in force output and disturbance resistance, but does not justify a system-level zero-power classification. Chen et al. describe dual ratchets for finer angular locking in a pneumatic hybrid gripper [15]. Their archival manuscript is discussed further in Section 5 because its energy comparison needs additional qualification.

4.2. Bistability and Contact-Triggered Capture

Bistability places retention within the structure's nonlinear mechanics. An external input moves the system across an energy barrier, after which stored energy drives motion toward another stable configuration. Chi et al. review the available structural and material approaches [9]. For grasping, the key issue is whether the loaded closed configuration preserves useful contact without unacceptable object motion.

A bistable dielectric elastomer gripper reported by Wang et al. uses a short voltage impulse and does not require continuous voltage to hold the object [30]. Its reported grasping-action energy is a useful device-level example, but it should not be generalized to a power-supply-inclusive pick-and-place cycle. Mechanical triggering offers a different route. Thuruthel et al. demonstrate rapid contact-triggered grasping and explicitly describe manual resetting in the inspected manuscript [18]. Passive closure therefore does not establish autonomous cyclic operation.

Lin et al. combine mechanical trigger sensitivity with pneumatic recharging in a bioinspired high-speed gripper [19]. The loaded state can persist without continuing external energy input, while the next cycle

requires restoration of the open state. This division reveals a general trade-off: reducing the trigger barrier improves sensitivity but can increase susceptibility to accidental activation. Increasing stored energy may strengthen or accelerate closure while raising contact-impact concerns.

Fast closure is only one part of dynamic capture. The finger-palm mechanism studied by Zhang et al. addresses energy harvesting and dissipation during target capture [31]. That emphasis complements bistable triggering because an object can rebound even when the fingers close quickly. The Fin Ray bistable gripper of Cai and Tang combines structural switching with adaptable finger-palm interaction [32]. Neither rapid response nor shape adaptation alone specifies the complete holding-and-release energy.

Structural customization can help align the switching trajectory with object requirements. He et al. use data-driven design of contact blocks to tailor bistable transition behavior and grasping forces [33]. The result supports design-specific tuning, rather than a universal superiority of bistability. Jin et al. further show how snapping metacaps can integrate rapid mechanical response and local sensing functions [34]. Their multiple robotic embodiments also underline the need to classify the particular gripper configuration rather than transferring resource claims between demonstrations.

Bistable fluidic architectures connect elastic stability to a closed fluid circuit. Qin et al. describe internal liquid redistribution between sensing and gripping chambers in a source-free gripper [14]. The configuration provides size-selective and stiffness-adaptive grasping without an external pump during the demonstrated operation. Its source-free designation concerns the demonstrated device boundary. Contact work, preparation, and repeatable release still need accounting before the entire robotic task can be called energy-free.

4.3. Jamming and Other Frictional Confinement

Jamming changes the ease of relative motion within a packed or layered structure. Brown et al. show that a granular gripper can combine friction, suction, and geometric interlocking at the object interface [20]. These contributions are not equivalent and vary with the object's surface and geometry. Amend et al. use positive pressure to improve release and reset in a granular gripper [35]. Release is therefore an active design variable within the jamming family.

The conventional description of granular, layer, and fibre jamming remains useful, but material arrangement alone does not identify the confinement source [10]. Caro and Carmichael review frictional and mechanical-interference mechanisms for laminar structures [11]. Their analysis cautions against equating all layer locking with continuously supplied vacuum. The resource condition must be determined from the actual normal-force mechanism, valves, and supply connections.

The distinction is particularly clear in passive particle jamming. Li et al. compress a particle pack through inflation of an adjacent soft actuator, avoiding a separate vacuum source [36]. The publisher-deposited abstract relates the stiffness increase to actuator air pressure. The term passive here describes the coupled stiffening process. It does not establish that the pressure can be removed while retaining the grasp.

Layer jamming can supply directional stiffness and shape constraints without requiring a bulky granular volume. Narang et al. model and characterize the transition from compliant to frictionally coupled laminates [21]. Ibrahimi et al. develop a layer-jamming actuator for tunable stiffness and shape change [37]. These studies inform load-path design, but a stiffness ratio is not itself a measurement of prolonged object retention.

Tensile jamming further separates resistance to stretching from resistance to bending. Yang et al. demonstrate segmented-fibril structures with controllable tensile stiffness and use them to modify soft-actuator trajectories [22]. This directional behavior can be valuable when a gripper needs to retain length while preserving another compliant motion. Its benefit depends on where the constrained direction lies within the grasp load path.

Frictional wrapping provides another route to load-dependent resistance. The coil-wrap gripper of Wei et al. uses frictional self-locking to enhance load capacity [38]. Such mechanisms may use the external load to

reinforce contact, but release and behavior under reversed loading remain essential questions. They should not be assigned a particular power-off capability from the title alone.

For pressure-confined jamming, stopping the pump and venting the chamber are fundamentally different interventions. An isolated chamber may retain vacuum for a finite period, whereas venting removes the confinement that supports friction. A low-leak circuit can consequently reduce maintenance power without changing the jamming material. Conversely, a material with high stiffness under supplied vacuum may lose useful retention rapidly after a seal fault. Current granular-gripper reviews discuss design and control across many implementations, reinforcing the need for circuit-specific comparisons [12].

4.4. Material Fixation and Thermal Switching

A stiff material state can preserve a conformed geometry after the shaping input changes. Low-melting-point alloys and shape memory polymers provide examples, but their switching and holding conditions differ. Hao et al. embed an alloy in a pneumatic finger, reshape it while molten, and allow it to recrystallize before loading [23]. The inspected paper distinguishes melting on a seconds scale from a cooling interval of several minutes.

This asymmetry affects task selection. Solidification can produce a useful holding state without continued heating, provided the operating environment maintains the required material state. Reconfiguration or release may then require reheating, and rapid repetition may require active cooling. The complete energy cost includes the thermal apparatus and waiting time. A material-level ability to retain shape does not establish economical or rapid object release.

Buckner et al. provide direct actuator-level evidence of this distinction [24]. Their pneumatic device is shaped while an electrically heated polymer composite is soft, held during six minutes of cooling, and then retains the actuated position after pneumatic pressure is released. Reheating permits recovery toward the initial shape. This supports material-state holding after pressure release, but it is not a gripper-payload test or a complete-cycle energy measurement.

The fully printable actuator of Liao et al. combines temperature-dependent stiffness with hydraulic regulation [25]. Its described apparatus includes heated and cooled reservoirs, pumps, and valves. This is a concrete example of why a compact finger can have a much larger system boundary. Shape-memory-alloy-driven grippers with variable stiffness provide another reminder that the actuation material and the retaining state must be identified separately [39]. The material name alone is insufficient to infer holding power.

Thermally switched adhesion also couples material state to contact mechanics. Son et al. use shape memory polymer stiffness control and a release-tip design to tune dry adhesion [27]. Heating and cooling affect both attachment and detachment, while conformal contact depends on preload and surface condition. Such systems broaden the possible grasp geometries, but their performance should be described through adhesion tests rather than treated as a finger-clamping payload.

4.5. Adhesive and Field-Based Boundary Cases

Some grippers retain objects mainly through interfacial traction. Shear-activated adhesive grippers demonstrate how an applied loading direction can engage a useful grasp without conventional squeezing [26]. Here the release trajectory and preload become part of the retention mechanism. Surface roughness, contamination, and object stiffness can change the result, so the object-interface conditions are indispensable comparison variables.

Combining material fixation with electroadhesion does not remove the electrical dependence of electroadhesion. In the low-melting-point-alloy electroadhesive gripper, the alloy is cooled and solidified before a high voltage is applied for adsorption [28]. The frozen geometry and the adhesive field perform different functions. A statement that heating can stop therefore does not imply that every grasp-supporting input can stop.

Electropermanent magnets offer a particularly instructive boundary test. The inspected study reports 50 mJ for switching the magnetic state and no power consumption by the magnet between switches [40]. However, its gripper demonstration uses a syringe pump providing a continuous flow of magnetorheological fluid. The switching-energy result belongs to the magnetic control element, not to the complete holding system. This distinction is central to interpreting low-power claims across hybrid architectures.

Fluidic intelligence should be assessed with the same care. The bistable valve of Rothemund et al. implements autonomous switching within a soft actuator circuit [41]. Passive reflex-response units similarly connect contact events to soft-robot reactions [42]. Removing electronic computation can simplify local control, yet continuing fluidic input may remain. Electronics-free behavior and source-independent retention are separate achievements.

5. Comparing Holding Performance and Energy

5.1. Preserve the Meaning of the Reported Quantity

Published load values can describe different mechanical questions. Maximum object mass measures successful retention under a particular grasp and motion. Pull-off force measures resistance along a specified extraction direction and often ends in failure. Blocked force characterizes an actuator rather than an object grasp. A stiffness ratio compares deformation responses under selected boundary conditions. These measurements should remain distinct, even when all are informally described as stronger gripping.

Table 3 provides quantitative anchors with their extraction limits. It is deliberately not a ranking. The examples establish different capabilities and measurement boundaries, and several do not report an energy quantity suitable for comparison. The evidence ledger supplies source locations and explains which conclusions follow from each observation.

Table 3. Selected quantitative anchors. Values are reported quantities, not harmonized benchmarks. Not established means the consulted source does not establish the stated energy or duration claim. No value in this table is interpreted as a transferable maximum rating.

Study and consulted source	Reported quantity and condition	Retention or release interpretation	Energy boundary and limitation
Brown et al. [20] Full text, Results and Discussion	About 75 kPa differential jamming pressure; bag radius 4.3 cm in the described tests.	Object retention combines friction, suction, and interlocking; vacuum provides confinement.	Pressure is not energy. Pump-off duration and source-inclusive hold energy are not established by these quantities.
Guo et al. [16] Publisher-deposited abstract	Load capacity up to 9 kg, as reported; a ratchet-pawl exoskeleton.	Holding without compressed gas; pneumatic unlocking.	A load-capacity statement and supply-off description, not an independently extracted endurance curve or full-cycle energy.
Wang et al. [30] Publisher abstract	0.04 s voltage impulse; 0.17 s grasping action; energy as low as 0.1386 J per grasping action.	Bistable dielectric elastomer gripper; no continuous voltage for holding.	Reported grasping-action energy. Inclusion of high-voltage conversion, release, reset, and long-duration holding is not established.
Thuruthel et al. [18] Inspected manuscript, experiments	Approximately 0.02 s sensing-and-closure response in the reported dynamic-grasp demonstration.	Stored elastic energy enables triggered closure; experiments use manual reset.	Response time is not cycle time; manual reset work and system cycle energy are not quantified here.
Hao et al. [23] Full text, Sections 2.4 and 3.1	Bottom-surface heating from 26 to 48 °C: about 37 s at 8 A and 160 s at 4 A; 3 min cooling before the solid-state pull-off test.	Pneumatic shaping followed by alloy solidification; reheating permits reshaping.	Surface-temperature criterion and cooling protocol are specified. These times are not a measured complete release cycle or a source-off retention-duration test.

Study and consulted source	Reported quantity and condition	Retention or release interpretation	Energy boundary and limitation
Electropermanent fluidic control [40] Full text, Section VI and Discussion	50 mJ magnetic-state switching; gripper demonstration uses 25 mL min−1 continuous flow and a 10 g object.	Magnet retains its field between pulses; pump supplies the demonstrated fluid circuit.	50 mJ is a component switching quantity. It does not establish zero system holding power.
Chen et al. [15] Archival manuscript, Section III-E	Abstract reports 42.6 J versus 85.28 J; body labels these totals with W and describes different transition/comparison windows.	Dual ratchets allow supply-off holding; servo releases the pawls.	Excluded from comparative energy calculations because units, equal-duration accounting, and boundary matching require clarification.

The last example illustrates why verification must extend beyond the abstract. In Chen et al., the stated actuation and reset powers and durations support a dimensional interpretation in joules [15]. However, repairing a unit label does not resolve whether all compared phases and holding windows are equivalent. We therefore retain the architecture as qualitative evidence and do not adopt the claimed percentage saving as a harmonized result.

5.2. Account for the Complete Grasp Cycle

A useful energy comparison declares its measurement boundary before its result. Let a complete cycle include closing, locking, holding, release, and reset. For non-overlapping phase intervals, the input energy at a declared electrical supply boundary is

$$E_{cycle} = E_{close} + E_{lock} + \int_0^{t_h} [P_{maint}(t) + P_{aux}(t)]\, dt + E_{release} + E_{reset} \quad (2)$$

The transition terms include every device inside that boundary during their respective phases. P_{maint} denotes the input required to sustain retention, and P_{aux} covers the other included loads during holding. Sensing, valve coils, control electronics, thermal equipment, and power converters should be counted when they lie inside the declared boundary. Overlapping actions must be integrated once rather than added twice.

Electrical measurements should use actual input power, including relevant conversion losses. For direct-current devices, time-resolved voltage and current can support the integral. Pneumatic pressure alone cannot establish consumed energy. A local hydraulic port can be approximated by pressure difference times volumetric flow, but pneumatic compression and expansion require a declared thermodynamic treatment. Upstream compressor or pump electricity should not be equated with ideal actuator-port work.

The same boundary issue applies to shared laboratory infrastructure. A comparison may report marginal compressed-air use, allocated compressor electricity, or total plant input. These answer different questions and need different accounting assumptions. Contact-triggered systems also receive mechanical work from the object, robot approach, or a person. That work should be reported separately when it is outside the electrical boundary, rather than silently assigned a value of zero.

The benefit of a lock depends on the holding interval. Under a simplified model with constant mean holding powers, define ΔE_{cycle} as locked-design energy minus matched-baseline energy. Let ΔE_{trans} be the locked design's extra transition energy and ΔP_{hold} the baseline holding power minus the locked design's holding power, all at the same boundary. Then

$$\Delta E_{cycle}(t_h) = \Delta E_{trans} - \Delta P_{hold}\, t_h, \quad t^* = \frac{\Delta E_{trans}}{\Delta P_{hold}} \quad (3)$$

A positive break-even time t* has this interpretation when both quantities are positive. If the transition cost is already lower, no positive minimum holding interval follows from this model. If holding power is not reduced, longer holding does not create the proposed benefit. Figure 2 shows the accounting logic without assigning synthetic device parameters or treating it as measured performance.

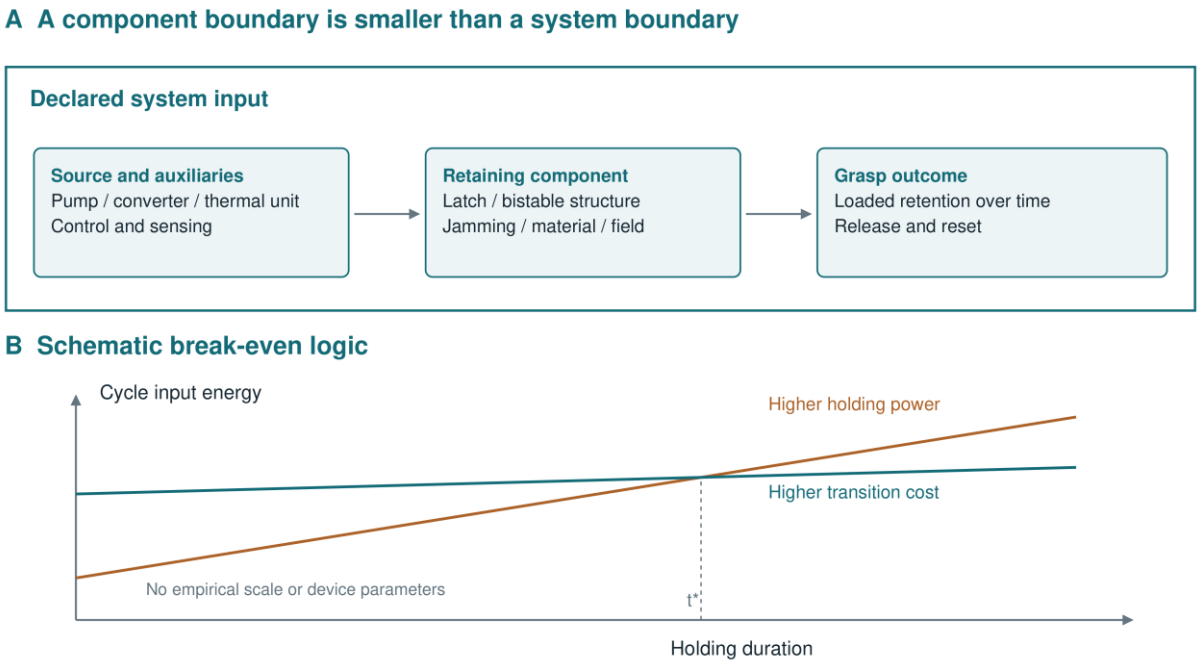


Figure 2. Symbolic energy accounting and break-even logic. Transition costs and holding-power differences are compared at the same system boundary. The intersecting curves are schematic and have no empirical scale or fitted parameters. A low-power retaining component can coexist with powered pumps, sensing, or thermal equipment.

5.3. Test Persistence Rather Than Inferring It

Isolated-pressure retention should include a pressure threshold. For illustration, assume an approximately isothermal, constant-volume chamber with a linearized leakage law, giving $\Delta p(t) = \Delta p_0 \exp(-t/\tau)$, where Δp_0 is the initial pressure difference and τ is the decay time constant. If Δp_{min} is the pressure difference required by a particular grasp, the predicted time t_p to reach that threshold is

$$t_p = \tau \ln\left(\frac{\Delta p_0}{\Delta p_{min}}\right), \quad for\ \Delta p_0 > \Delta p_{min} \tag{4}$$

Equation (4) follows from an assumed decay law; it is not a fit to a reviewed gripper or a guaranteed object-retention time. Slip, force relaxation, or contact loss can end a grasp before the pressure threshold is reached. Real chambers may also change volume, temperature, and permeability. Pressure and loaded retention must therefore be measured together before the model can predict a useful holding interval.

Mechanical locks have a different persistence problem. Fixed finger position does not prevent viscoelastic relaxation, object deformation, or loss of contact area. A retention test should therefore monitor object motion and, where feasible, force or another validated proxy. A source-off test with no external load cannot establish loaded retention. Similarly, surviving one interrupted cycle provides weaker endurance evidence than repeated holds across relevant objects and disturbance directions.

6. Release, Selection, and Control

6.1. Treat Release as a Required Function

The mechanism that most effectively resists reopening can also make commanded release difficult. Ratchet disengagement may have to overcome contact loading; a bistable gripper must cross a reverse barrier; a jammed gripper must lose sufficient confinement. A fixed material state may require reheating, while an adhesive interface may require peeling or unloading. These are consequences of the respective load paths, and they motivate testing release under the intended retained load.

Three release times should be distinguished: time to initiate unlocking, time to clear the object, and time to restore readiness for the next grasp. A fast snap-through event can coexist with slow reset. The manual reset

described by Thuruthel et al. and the thermal waiting interval in Hao et al. illustrate different sources of this asymmetry [18, 23]. Their times should not be collapsed into a single actuator-speed comparison.

Release should also be evaluated when the normal supporting source is unavailable. A valve-controlled unlock may depend on pressure that has been lost, and an electrically actuated pawl may be inaccessible after battery depletion. These are design scenarios requiring verification, rather than claims that a particular reviewed prototype necessarily fails. A useful specification states both the desired loss-of-power state and the available recovery path.

6.2. Select Mechanisms from the Task Boundary

Long stationary holds favor investigation of mechanisms that reduce maintenance input, provided engagement and release remain reliable. A ratchet or an adequate loaded bistable state can meet this objective without maintaining a fluidic resource. An isolated jamming circuit may also be suitable when leakage over the required interval is characterized. The preferred architecture depends on the object, disturbances, acceptable drift, and the energy boundary.

Rapid repetitive manipulation makes transition energy and reset time more influential. A thermal fixation mechanism may be attractive for an infrequently reconfigured fixture yet poorly matched to a short-cycle sorting task. Conversely, a supply-sustained gripper can remain competitive when holds are brief and switching a separate lock adds substantial cost. Equation (3) provides a conditional selection test, not a technology ranking.

Fragile-object tasks require evidence beyond maximum payload. Lock engagement, snap-through, and pressure changes can alter peak contact forces during acquisition. Subsequent force relaxation may reduce damage but also reduce slip resistance. Designs that decouple soft contact from a supporting skeleton offer one approach, while tailored bistable transitions offer another [16, 33]. Both need object-specific contact and release assessment.

Unstructured and underwater tasks require additional attention to the environment. Ambient pressure changes the available pressure differential, and moisture can alter interfacial behavior. Source-free fluidic designs and adaptable bistable grippers are relevant examples, but their demonstrations do not define universal environmental limits [14, 32]. Figure 3 expresses mechanism selection as a sequence of evidence questions rather than a scored comparison with unsupported weights.

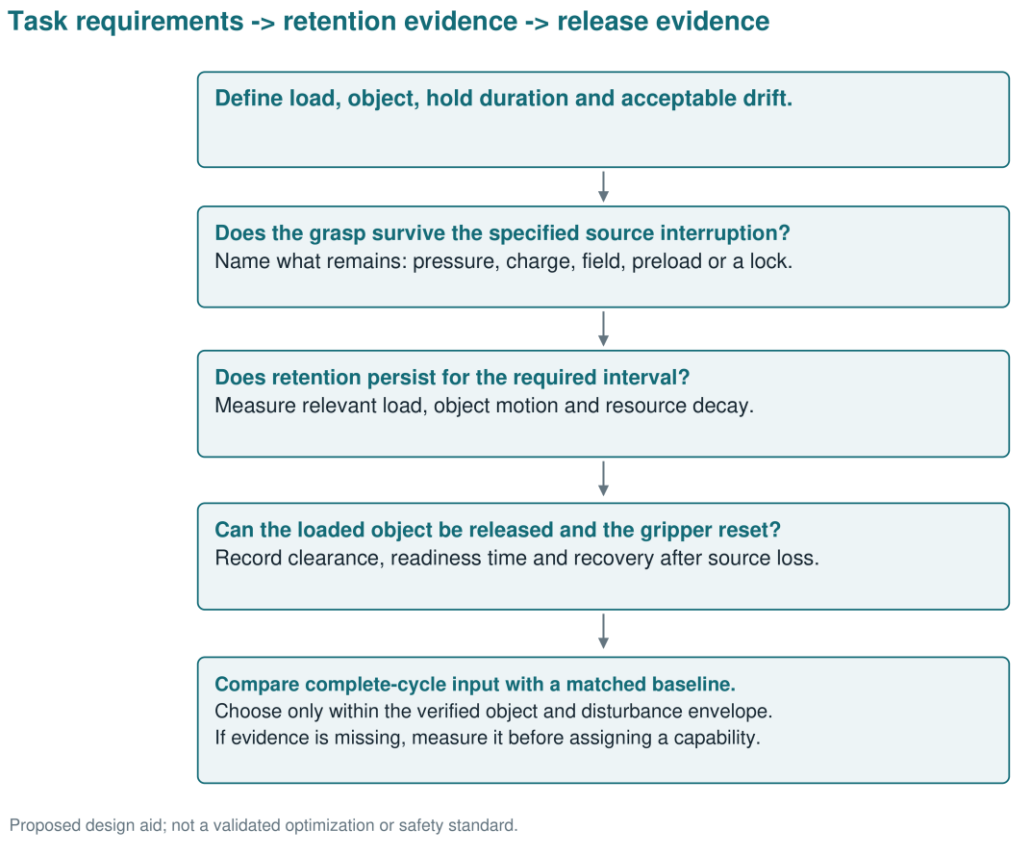


Figure 3. A proposed decision sequence for mechanism selection. Each branch asks for a task requirement or a measurable capability. The sequence is a design aid and has not been validated as an optimization algorithm. Inconclusive source-off evidence should lead to measurement, rather than an assumed resource classification.

6.3. Connect Mechanical Retention to Embodied Control

Contact-triggered grippers show that local mechanical interactions can perform part of the sensing-to-action transition [18, 19, 34]. This is relevant to embodied manipulation because physical design can reduce the need for a separately computed closure event. The remaining supervisory problem is to determine whether acquisition succeeded, retention persists, and release is complete. Mechanical intelligence changes those control requirements rather than eliminating them.

A possible controller could use acquisition, engagement verification, holding, release verification, and recovery as explicit operating states. Transitions would depend on measured contact, latch position, pressure, or slip indicators as appropriate to the mechanism. For example, a pressure trace after confirmed circuit isolation could track the trapped resource, while a separate contact measurement could detect grasp loss through object deformation. This architecture is a proposal derived from the review; its performance has not been experimentally tested here.

Learning-based design and control should preserve the same resource and release boundaries. Data-driven bistable design already provides an example of mapping geometry to transition behavior [33]. Extending such approaches to retention duration or complete-cycle energy requires corresponding labels and measurements. Optimizing a closure success rate alone would leave the holding and release objectives unspecified.

These considerations also qualify the meaning of autonomous grasping. Autonomous triggering can be demonstrated without autonomous reset. Electronics-free switching can be demonstrated with a continuing pressure source. Low coil power can be demonstrated with a powered fluid supply [40, 41, 42]. Reporting these distinctions gives a controller designer more actionable information than an unqualified low-power label.

7. Reporting Priorities and Open Questions

The common conclusion concerns how retention evidence should be interpreted. Retention, supply dependence, and complete-cycle energy are distinct properties and require distinct measurements. Existing jamming reviews already identify inconsistent performance evaluation as an obstacle [10, 11]. The reporting set in Table 4 extends that concern to source interruption, holding duration, and release across mechanism families.

Table 4. Proposed minimum reporting set for locking and low energy holding studies. These are review recommendations, not an existing journal or community standard.

Item	Minimum information	Interpretation enabled
Task and object	Object dimensions, mass, surface, compliance, grasp orientation, and robot motion.	Distinguishes contact-dependent capacity from an actuator rating.
Load path	Retention element, constrained variable, supporting contacts, and relevant load direction.	Shows whether locking position also supports the grasp.
Resource circuit	Pump, valves, heaters, coolers, converters, fields, controller, and sensor connections.	Defines what remains connected during holding.
Interruption test	Precisely state what is switched off, disconnected, vented, or left isolated.	Distinguishes supply-off operation from resource removal.
Holding interval	Load, duration, drift criterion, force or pressure history, and failure endpoint.	Supports a bounded persistence claim.
Acquisition and trigger	Preload, contact speed, peak contact force where relevant, and trigger threshold.	Connects capture reliability to object loading and false activation.
Release and reset	Command, retained load, time to clearance, time to readiness, and energy or external work.	Separates fast unlocking from a repeatable complete cycle.
Energy measurement	Supply boundary, all included components, sampling method, phase intervals, and infrastructure allocation.	Allows matched energy comparisons and avoids component-to-system inference.
Repeatability	Trial count, object variation, wear or fatigue exposure, failures, and relevant environmental conditions.	Bounds reliability and transfer to another task.
Matched baseline	Same object, orientation, hold duration, success criterion, and energy boundary.	Tests whether lower holding input yields a lower task-level cost.

7.1. Which Locks Preserve Contact Force over Time?

A mechanically retained joint can coexist with a compliant and relaxing contact. The unresolved question is how geometry, object compliance, and material relaxation determine the duration of useful force. A discriminating experiment would compare locked position, contact force, and slip under the same load across several holding intervals. Its outcome would show when a position lock is sufficient and when additional force regulation or mechanical compensation is needed.

Such tests should separate adaptation benefits from retention deficits. A larger contact area may improve tolerance to object variation while a softer contact loses force more rapidly. Neither tendency can be inferred reliably from a single successful lift. Time-resolved measurements would connect soft-contact design to the actual duration requirement, rather than treating compliance as an unconditional advantage or disadvantage.

7.2. When Does Passive Holding Reduce Task Energy?

The second open question is whether reduced maintenance input survives complete-cycle accounting. A useful comparison would vary holding duration while keeping object, motion, and energy boundary fixed. Closing, engagement, holding, release, and reset would all be measured for the locked and baseline designs.

The resulting curves would test the assumptions behind Equation (3) and reveal whether a practical break-even interval exists.

For fluidic systems, the test should distinguish component electricity from upstream source input. The continuous-flow electropermanent demonstration shows why this distinction can change the interpretation of an otherwise valid low-power component result [40]. For thermal systems, reservoir preparation and cooling should be included or explicitly amortized. These measurements would make application-level recommendations more defensible than comparisons of selected switching energies.

7.3. Can Release Remain Reliable as Retention Strength Increases?

Stronger retention can increase the work or control authority required for release. A useful investigation would measure release probability, time, and input under increasing retained load and relevant disturbances. It should include realistic unloading opportunities and a stated power-loss condition. Such evidence would reveal whether improved holding capacity translates into usable manipulation or creates a release bottleneck.

Release failures also affect energy comparisons. Repeated attempts, recovery motions, or discarded objects can outweigh savings measured during a successful hold. Reporting both successful-cycle energy and failure frequency would connect mechanism design to task reliability. The reviewed literature supports the importance of release mechanisms, but the consulted corpus does not support a pooled numerical relation between holding strength and release reliability.

7.4. Limits of the Present Synthesis

The targeted search favors accessible English-language publications, known mechanism terms, and work indexed by the consulted services. It cannot establish exhaustive coverage or an unbiased estimate of technology prevalence. Some retained studies were available only through abstracts, limiting extraction of circuit details and test conditions. The evidence ledger makes these restrictions visible so that interpretation does not silently exceed source access.

The proposed classification and reporting set have not been validated through inter-reviewer coding or a prospective benchmark. Hybrid devices can occupy several mechanism families, and their resource condition can change with the circuit state. We therefore present the framework as an explicit comparison aid, rather than a standardized taxonomy. Its value depends on preserving the physical load path and the conditions attached to each evidence statement.

No new experiments, digitized force curves, or pooled effect estimates are introduced. Short demonstrations cannot be converted into lifetime, deployment readiness, or safety guarantees. The review identifies supported conclusions and the measurements needed to resolve the remaining uncertainties. Broader claims about industrial autonomy or embodied intelligence require task-level evidence beyond the present review.

8. Conclusions

Locking and low energy holding should be compared through the mechanism that retains the load and the resources that remain necessary. Mechanical engagement, elastic bistability, frictional confinement, material fixation, and interfacial traction provide distinct routes with different release requirements. A mechanism name alone does not establish power-off object retention or low complete-cycle energy.

The two-axis framework connects these routes to sustained supplies, isolated resources, and state-retained operating conditions. It also separates position retention from force persistence, and component switching energy from system input. Representative studies support these distinctions, while unmatched tests and incomplete energy boundaries prevent a defensible universal ranking.

Mechanism selection should therefore begin with the holding interval, load path, acceptable drift, and available release authority. Matched cycle measurements and loaded interruption tests are the next steps needed to turn promising retention concepts into transferable design evidence. For robotic manipulation, the objective is a repeatable acquisition-hold-release cycle with a verified resource cost.

Supplementary Materials

Supplementary Material S1 contains the evidence ledger, search approach, access limitations, and source-verification notes. Machine-readable extraction records and bibliographic files accompany the submission package.

Author Contributions

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Institutional Review Board Statement

Not applicable. This review introduces no new studies involving humans or animals.

Informed Consent Statement

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Data Availability Statement

All evidence used in this review is identified in the reference list and Supplementary Material S1. The extraction ledger and search record accompany the manuscript. No new experimental data were generated.

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Conflicts of Interest

[AUTHOR TO COMPLETE: declare all actual conflicts, or confirm that there are none.]

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